GEK-83838A

DRIVE SYSTEM INSTRUCTIONS

# INTERPRETING T-GSI TAPE DATA

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC

# INTERPRETING

# T-GSI

# TAPE DATA

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## NOTES

## INTRODUCTION

This publication describes the format used by the T-GSI system for recording data on magnetic tape. The tape consists of two major parts - HEADER, and RAW DATA RECORDS (LOGS).

The HEADER is provided as a means by which the user may identify and interpret the Raw Data using his computer. This may be done by linking the HEADER with the users program for reading and manipulating the tape data.

## DESCRIPTION

The T-GSI system's historical data is permanently stored on 9-track magnetic tape using an IBM compatible phase encoded recording configuration. (Refer to the recorder instruction book.)

The recording density is 1600 CPI. Record length is 2048 bytes.

## RECORDING FORMAT

The recording format of the tape is as follows:

Header

End Of File mark

1st day's Logs (2048 bytes each)

End Of File mark

2nd day's Logs

End Of File mark

3rd etc

Note: If a partially full tape is removed from the the T-GSI system, five End Of File marks are recorded after the last log.

## HEADER

### GENERAL

All information needed to format the T-GSI tape data can be obtained from the Identification and Raw Data Descriptor Databases contained in the HEADER, recorded at the beginning of each T-GSI tape.

### IDENTIFICATION DATABASE

The first eight records and/or until the word FFFFH is detected, is the Identification database file.

This file consist of one entry per T-GSI channel used in the system as follows:

- channel number (1 word)
- 20 character HEX-ASCII unique identifier (20 bytes)
- units code (1 byte)

The following is a listing of units codes used to date:

NOTE: Additional units code may appear on some systems. These may be reconciled by matching codes against unit given for that channel number on the Customer Connection Diagram for the particular system in question.

ANALOG CHANNELS

| UNITS CODE | UNITS |
|---|---|
| 00H | MILS |
| 01H | INCHES |
| 02H | RPM |
| 03H | PERCENT |
| 04H | PSIA |
| 05H | PSIG |
| 06H | DEG F |
| 07H | MW |
| 08H | MVAR |
| 09H | KA |
| 0AH | AMPS |
| 0BH | KV |
| 0CH | VOLTS |

## HEADER (Continued)

ANALOG CHANNELS (Continued)

| UNITS CODE | UNITS |
|---|---|
| 0DH | HZ |
| 0EH | UM/CM |
| 0FH | GPM |
| 10H | DEG C |
| 11H | CHN NO |
| 12H | MA |
| 13H | " "(NO UNITS) |

DIGITAL CHANNELS

| UNITS CODE | 0-NORMAL | 1-ABNORMAL |
|---|---|---|
| 80H | AC | DC |
| 81H | TRIP | CLOSED |
| 82H | NORMAL | HIGH |
| 83H | NORMAL | LOW |
| 84H | NORMAL | FULL |
| 85H | NORMAL | V HIGH |
| 86H | RNING | OFF |
| 87H | COMP | OPER |
| 88H | DISENG | ENGAGE |
| 89H | TURN | STOP |
| 8AH | NORMAL | FLRE |
| 8BH | NORMAL | OVLD |
| 8CH | NORMAL | OVLTGE |
| 8DH | NORMAL | OVCRNT |
| 8EH | NORMAL | ALARM |
| 8FH | NORMAL | FL REV |
| 90H | NORMAL | WT DET |
| 91H | NO | YES |
| 92H | NORMAL | LCKOUT |
| 93H | LAG | LEAD |
| 94H | NORMAL | DEM CL |
| 95H | NORMAL | FULL |
| 96H | ON | OFF |
| 97H | YES | NO |
| 98H | OFF | RNING |
| 99H | NORMAL | TRIP |
| 9AH | NORMAL | ALO |
| 9BH | NORMAL | FAIL |
| 9CH | STOP | RUN |

## HEADER (Continued)

DIGITAL CHANNELS Continued)

| UNITS CODE | 0-NORMAL | 1-ABNORMAL |
|---|---|---|
| 9DH | NORMAL | RUN |
| 9EH | OPEN | CLOSED |
| 9FH | OPER | COMPUT |
| A0H | CLOSED | OPEN |

RAW DATA DESCRIPTOR DATABASE

Immediately following the Identification Database file is the Raw Data Descriptor database file. The Raw database file occupies four records on tape and consists of an index table and several data Record tables.

The first few words in the file is the index table which points to the different data Record tables used by T-GSI. The only data Record table of concern here is the first table, which is the Raw Data Descriptor Records (described below). Therefore, use the first word in the file to determine the number of bytes to shift down to the beginning of the Raw Data Descriptor Record table. The last Record in this table has FFFFH as the channel number and all zero data.

RAW DATA RECORD

The Raw Data Descriptor table consist of one Record for each group of common channels. Each Raw Data Descriptor Record is a fixed length (16 bytes) Record. The order of the Records directly correspond to the order of the channels in the system log.

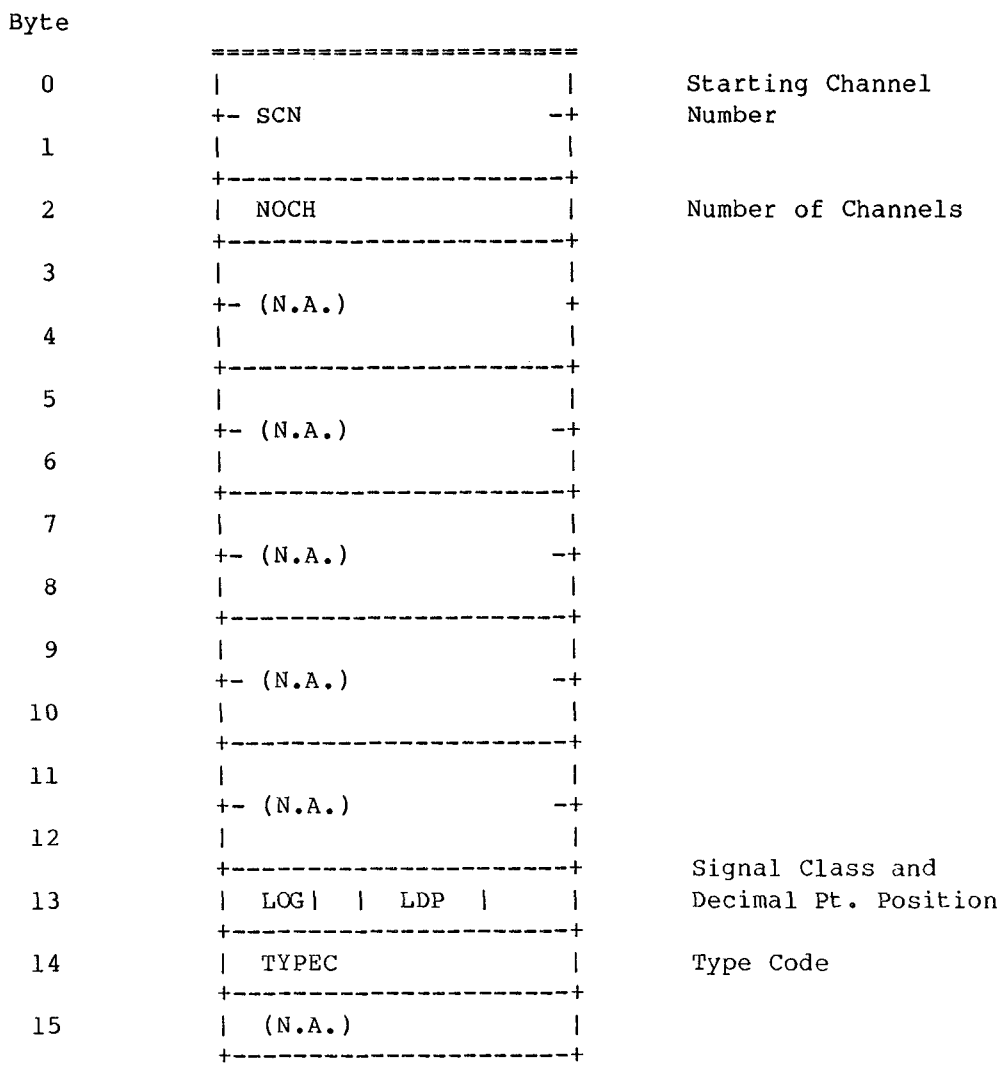
NOTE: Unused channels are recorded as zeroes.

There are five fields of interest when dumping data. They are:

| | |
|---|---|
| SCN | Starting Channel Number |
| NOCH | Number of Channels |
| LOG | Signal Class |
| LDP | Number of Digits to the Left of the Decimal Point. |
| TYPEC | Input Type Code |

The layout of one Raw Data Descriptor Record is illustrated on the following page.

HEADER (Continued)




The Raw Data Descriptor fields are defined below.

| Field | Byte | Bits | Description |
|---|---|---|---|
| SCN | 0 | 7:0 | Starting Channel Number - Low Half. |
| | 1 | 7:0 | Starting Channel Number - High Half. |
| NOCH | 2 | 7:0 | Number of Channels - Starting with SCN |
| LOG | 13 | 7 | Signal Class (0 = Analog: 1 = Digital) |
| LDP | 13 | 5:3 | Number Digits to Left of Decimal Point |
| TYPEC | 14 | 7:0 | Input Type Code |

**HEADER** (Continued)

Type Codes of Interest

- C0H = Digital input (contact)
- Any type code with an associated Unit Code of 11H (See Identification Database). These are software points that return channel numbers.
- All others are don't care.

As an example, the following Raw Data Descriptor Record defines channels 11 to 19 as analog channels where there are two decimal digits to the left of the decimal point.

| Byte | Value | Comments |
|---|---|---|
| 0 | 0BH | Starting channel is 11. |
| 1 | 00H | |
| 2 | 09H | There are 9 channels. |
| 3 | XX | Note: XX = Don't Care |
| 4 | XX | |
| 5 | XX | |
| 6 | XX | |
| 7 | XX | |
| 8 | XX | |
| 9 | XX | |
| 10 | XX | |
| 11 | XX | |
| 12 | XX | |
| 13 | 10H | NOTE: Bits 0:2 and bit 6 |
| 14 | 57H | are don't care. |
| 15 | XX | |

The twentieth and subsequent channel values must be interpreted using the next successive Raw Data Descriptor Field.

The end of the Raw Data database file which is also the end of the HEADER, is followed by a End Of File mark.

## RAW DATA

### LOG MAKE-UP

The make-up of each log on tape is as follows: (2048 bytes)

Note: Each point occupies two bytes.

| Point No. | Description |
|---|---|
| 0 | Year |
| 1 | Month |
| 2 | Day |
| 3 | Hours(24 hr. clock) |
| 4 | Minutes |
| 5 | Seconds |
| 6 | TN1 (Turbine Number - 4 words) |
| 7 | TN2 |
| 8 | TN3 |
| 9 | TN4 |
| 10 | unused |
| 11 to 1024 | Data: Refer to the Customer Connection diagram for detailed listing. Unused points are recorded as zeroes. |

### LOG DATE

The Log Date is a three word hexadecimal string defining the date when the log was taken. The format of the date is "Y2Y1/M2M1/D2D1".

### LOG TIME

The Log Time is a three word hexadecimal string defining the time when the log was taken. The format of the time is "h2h1:m2m1:s2s1". Time is given in terms of a 24 hour clock.

### TURBINE NUMBER

The Turbine Number is a four word HEX-ASCII string corresponding to the system turbine number.

**RAW DATA (Continued)**

| | |
|---|---|
| Y2 Y2 Y2 Y2 Y2 Y2 Y2 Y2 | Date of Log |
| Y1 Y1 Y1 Y1 Y1 Y1 Y1 Y1 | |
| M2 M2 M2 M2 M2 M2 M2 M2 | |
| M1 M1 M1 M1 M1 M1 M1 M1 | |
| D2 D2 D2 D2 D2 D2 D2 D2 | |
| D1 D1 D1 D1 D1 D1 D1 D1 | |
| h2 h2 h2 h2 h2 h2 h2 h2 | Time of Log |
| h1 h1 h1 h1 h1 h1 h1 h1 | |
| m2 m2 m2 m2 m2 m2 m2 m2 | |
| m1 m1 m1 m1 m1 m1 m1 m1 | |
| s2 s2 s2 s2 s2 s2 s2 s2 | |
| s1 s1 s1 s1 s1 s1 s1 s1 | |
| TN1 - low byte | Turbine Number |
| TN1 - high byte | |
| TN2 - low byte | |
| TN2 - high byte | |
| TN3 - low byte | |
| TN3 - high byte | |
| TN4 - low byte | |
| TN4 - high byte | |
| 1st data point-low byte | |
| 1st data pt.- high byte | |
| 2nd data pt. etc | |

Figure 1 Tape Log format

## RAW DATA (Continued)

### CHANNEL DATA

Channel data is recorded in one of three formats, depending on whether it represents analog or digital data, or whether it is a T-GSI software point that returns a channel number.

Analog Input

| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| signed two's complement representation | | | | | | | | | | | | | | S | S |

Digital Input

| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| all 1's = Abnormal or all 0's = Normal | | | | | | | | | | | | | | S | S |

Normal, Abnormal - See Digital Units Code table

Software Channel No.

| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Binary Channel number | | | | | | | | | | | | | | S | S |

| S | S | Description |
|---|---|---|
| 0 | 0 | Normal |
| 0 | 1 | Pre-alarm condition |
| 1 | 0 | Alarm condition |
| 1 | 1 | Failed channel |

As shown above, the channel value comprises two fields - a value field (bits 2 - 16) and a status field (bits 0 and 1).

The channel value is a 14-bit, signed two's complement representation (analog channels), all 1's or 0's for digital channels, or straight binary if the point is a channel number.

The two-bit status field defines the operational status of the channel.

## RAW DATA (Continued)

CHANNEL DATA (Continued)

The following examples illustrate how the T-GSI system records the channel value and status. As an example, the T-GSI system will record the hexadecimal string "1236" if the channel is in an alarm state and the channel value is 1165D (48DH).

| 14-bit Channel Value | | | | Status |
|---|---|---|---|---|
| 4 | 8 | D | | |
| 0 0 0 1 0 0 1 0 | 0 0 0 1 1 0 1 | | | 1 0 |
| 1 | 2 | 3 | 6 | |
| 16-bit Representation | | | | |

If the channel value is -1165D (3B73H) and in an alarm condition, the hexadecimal string "EDCE" would be recorded.

| 14-bit Channel Value | | | | Status |
|---|---|---|---|---|
| 3 | B | 7 | 3 | |
| 1 1 1 0 1 1 0 1 | 1 1 1 0 0 1 1 | | | 1 0 |
| E | D | C | E | |
| 16-bit Representation | | | | |

If the channel is a digital input and in the abnormal state with no alarm the hexadecimal string "FFFC" would be recorded.

| 14-bit Channel Value | | | | Status |
|---|---|---|---|---|
| 3 | F | F | F | |
| 1 1 1 1 1 1 1 1 | 1 1 1 1 1 1 1 | | | 0 0 |
| F | F | F | C | |
| 16-bit Representation | | | | |

## RAW DATA (Continued)

CHANNEL DATA (Continued)

If the channel is a software point returning the channel number 796D(31CH) with no alarm, the hexadecimal string 0C70H would be recorded.

| | | | | | | | | | | | | | | | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 14-bit Channel Value | | | | | | | | | | | | | | | |
| | | 3 | | | | 1 | | | | C | | | | | |
| 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| | 0 | | | C | | | 7 | | | 0 | | | | | |
| 16-bit Representation | | | | | | | | | | | | | | | |

It should be noted that channel numbers, signal polarity, and decimal point position are not included with the value. It is the responsibility of the users computer to format the channel values using the information provided in the tape HEADER.

NOTES

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